```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
np.random.seed(10)
students = ['S' + str(i) for i in range(1, 21)]
df = pd.DataFrame({
 'Name': students,
 'Math': np.random.randint(50, 100, 20),
 'Science': np.random.randint(50, 100, 20),
 'English': np.random.randint(50, 100, 20)
 })

df['Total'] = df[['Math', 'Science', 'English']].sum(axis=1)
df['Average'] = df['Total'] / 3
df['Grade'] = pd.cut(df['Average'],
bins=[0, 60, 75, 90, 100],
labels=['C', 'B', 'A', 'A+'])
df = df.sort_values(by='Total', ascending=False)

print("Sample Data:\n")
print(df.head())
fig = plt.figure(figsize=(12, 8))

# KPI Section
top_score = df['Total'].max()
avg_score = df['Total'].mean()

plt.subplot(2, 2, 1)
plt.text(0.5, 0.6, f"Top Score\n{top_score}", fontsize=14, ha='center')
plt.text(0.5, 0.3, f"Average Score\n{round(avg_score, 2)}", fontsize=14, ha='center')
plt.title("Key Metrics")
plt.axis('off')

# Subject Average Bar Chart
plt.subplot(2, 2, 2)
subject_avg = df[['Math', 'Science', 'English']].mean()
subject_avg.plot(kind='bar')
plt.title("Subject Average Marks")
plt.ylabel("Marks")

# Student Performance Line Chart
plt.subplot(2, 2, 3)
plt.plot(df['Name'], df['Total'], marker='o')
plt.xticks(rotation=45)
plt.title("Student Total Marks")
plt.xlabel("Students")
plt.ylabel("Total Marks")

# Distribution Histogram
plt.subplot(2, 2, 4)
plt.hist(df['Total'])
plt.title("Marks Distribution")
plt.xlabel("Total Marks")
plt.ylabel("Count")

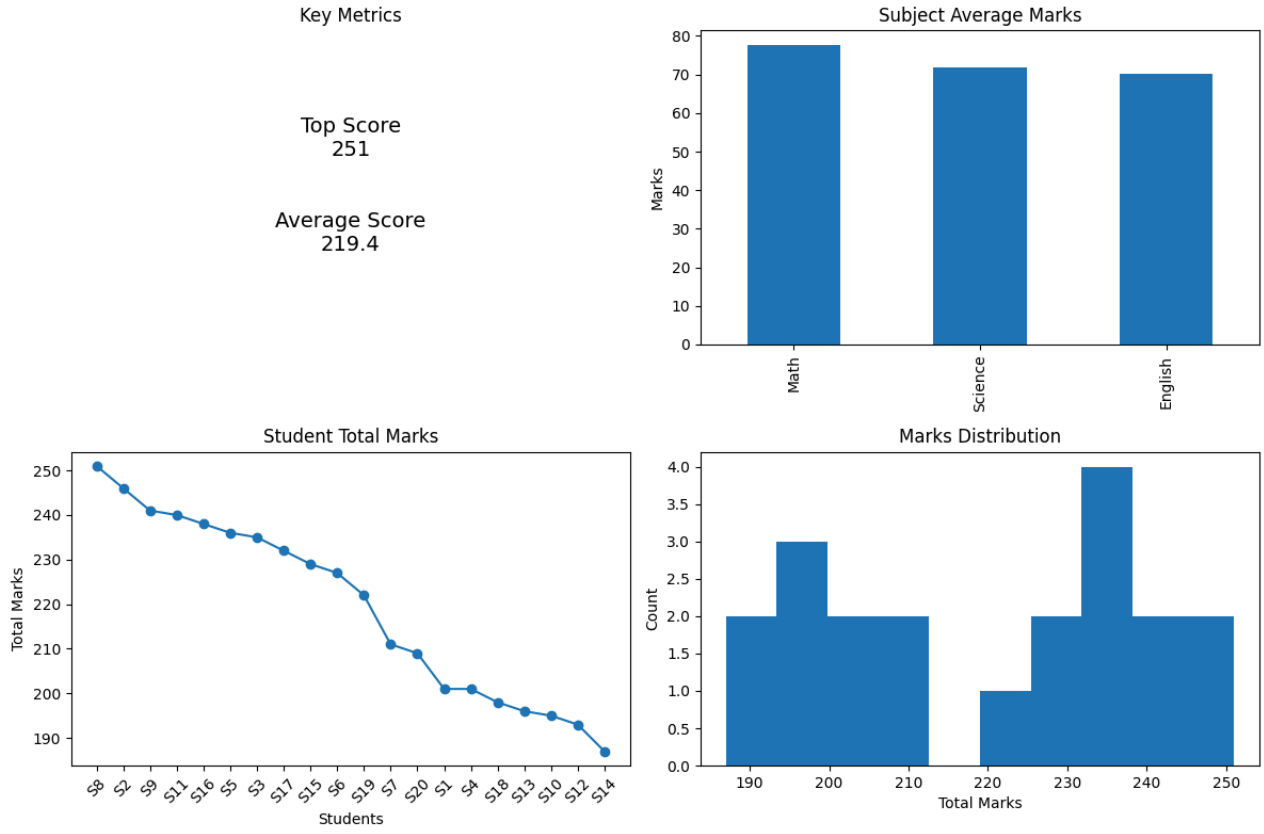
# Final Layout
plt.tight_layout()

# Save as Image (optional)
plt.savefig("student_dashboard.png")

# Show Dashboard
plt.show()
```

```
Sample Data:

   Name  Math  Science  English  Total    Average Grade
7    S8    79       99       73    251  83.666667     A
1    S2    86       74       86    246  82.000000     A
8    S9    98       63       80    241  80.333333     A
10  S11    99       63       78    240  80.000000     A
15  S16    90       80       68    238  79.333333     A
```




```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

np.random.seed(3)

patients = [f'P{i}' for i in range(1,21)]
departments = ['Cardio','Neuro','Ortho','General']

df = pd.DataFrame({
 'Patient': patients,
 'Department': np.random.choice(departments,20),
 'Age': np.random.randint(20,70,20),
 'Bill': np.random.randint(2000,10000,20)
 })

total_patients = len(df)
avg_bill = df['Bill'].mean()
dept_count = df['Department'].value_counts()

plt.style.use('ggplot')
fig = plt.figure(figsize=(14,8))

# KPI
plt.subplot(2,2,1)
plt.text(0.5,0.6,f"{total_patients}",ha='center',fontsize=22)
plt.text(0.5,0.3,f"Avg Bill ₹{round(avg_bill,2)}",ha='center')
plt.title("Hospital KPI")
plt.axis('off')

# Dept count
plt.subplot(2,2,2)
dept_count.plot(kind='bar')
plt.title("Patients by Dept")
```

```
# Bill trend
plt.subplot(2,2,3)
plt.plot(df['Patient'],df['Bill'],marker='o')
plt.xticks(rotation=45)
plt.title("Patient Bills")

# Pie
plt.subplot(2,2,4)
dept_count.plot(kind='pie', autopct='%1.1f%%')
plt.title("Dept Share")

plt.tight_layout()
plt.show()
```




```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

np.random.seed(2)

movies = [f'M{i}' for i in range(1,21)]
genres = ['Action','Comedy','Drama','Horror']

df = pd.DataFrame({
'Movie': movies,
'Genre': np.random.choice(genres,20),
'Rating': np.round(np.random.uniform(5,10,20),1)
})

avg_rating = df['Rating'].mean()
top_movie = df.loc[df['Rating'].idxmax(),'Movie']
genre_avg = df.groupby('Genre')['Rating'].mean()

plt.style.use('fivethirtyeight')
fig = plt.figure(figsize=(14,8))

# KPI
plt.subplot(2,2,1)
plt.text(0.5,0.6,f"{round(avg_rating,2)}⭐",ha='center',fontsize=22)
plt.text(0.5,0.3,f"Top: {top_movie}",ha='center',fontsize=12)
plt.title("Movie KPI")
plt.axis('off')

# Genre bar
plt.subplot(2,2,2)
genre_avg.plot(kind='bar')
plt.title("Genre Ratings")

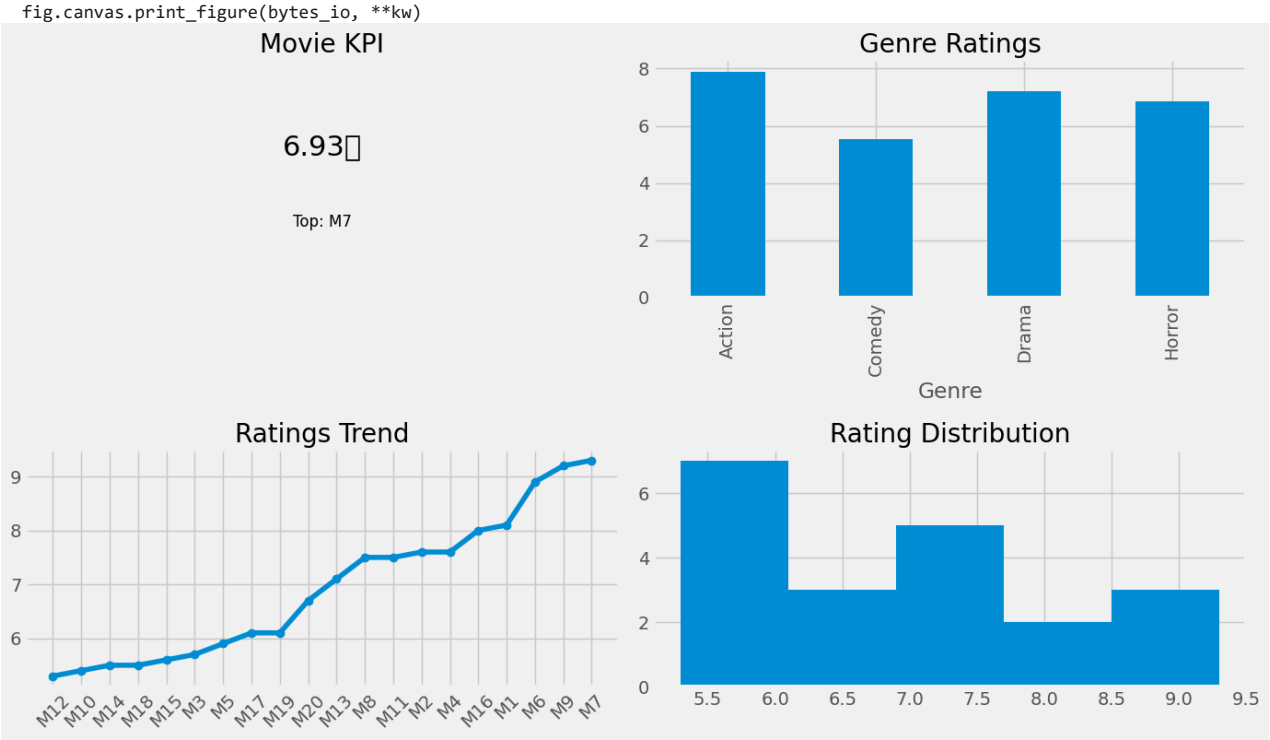
# Trend
```

```
# Trend
plt.subplot(2,2,3)
df_sorted = df.sort_values('Rating')
plt.plot(df_sorted['Movie'],df_sorted['Rating'],marker='o')
plt.xticks(rotation=45)
plt.title("Ratings Trend")

# Histogram
plt.subplot(2,2,4)
plt.hist(df['Rating'],bins=5)
plt.title("Rating Distribution")

plt.tight_layout()
plt.show()
```

```
/tmp/ipykernel_807/1212828333.py:47: UserWarning: Glyph 11088 (\N{WHITE MEDIUM STAR}) missing from font(s) DejaVu Sans.
  plt.tight_layout()
/usr/local/lib/python3.12/dist-packages/IPython/core/pylabtools.py:151: UserWarning: Glyph 11088 (\N{WHITE MEDIUM STAR}) mis
  fig.canvas.print_figure(bytes_io, **kw)
```




```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

np.random.seed(1)

days = [f'D{i}' for i in range(1, 21)]
products = ['Laptop', 'Phone', 'Tablet', 'Accessories']

df = pd.DataFrame({
'Day': days,
'Product': np.random.choice(products, 20),
'Sales': np.random.randint(1000, 5000, 20)
})

total_sales = df['Sales'].sum()
avg_sales = df['Sales'].mean()
product_sales = df.groupby('Product')['Sales'].sum()

plt.style.use('bmh')
fig = plt.figure(figsize=(14, 8))

# KPI
plt.subplot(2,2,1)
plt.text(0.5,0.6,f"₹{total_sales}",ha='center',fontsize=22,weight='bold')
plt.text(0.5,0.3,f"Avg ₹{round(avg_sales,2)}",ha='center',fontsize=14)
plt.title("Sales KPI")
plt.axis('off')

# Bar
```

```
plt.subplot(2,2,2)
product_sales.plot(kind='bar')
plt.title("Product Sales")

# Line + fill
plt.subplot(2,2,3)
plt.plot(df['Day'],df['Sales'],marker='o')
plt.fill_between(df['Day'],df['Sales'],alpha=0.2)
plt.xticks(rotation=45)
plt.title("Trend")

# Pie (attractive!)
plt.subplot(2,2,4)
product_sales.plot(kind='pie', autopct='%1.1f%%')
plt.title("Sales Share")

plt.tight_layout()
plt.show()
```

